

In this project, I went line by line with specific intentions. I started with `iostream`, `namespace std`, the classics, `int main()`, etc. but also made sure to include `<vector>`, `<string>`, `<cctype>` (for `isalpha()` and `tolower()`), and `<algorithm>` (for `sort()`). I started with the function `cleanWord`, which creates an empty string and uses a range based loop to clean the string character by character. My next function, `getWords`, reads words from the file, uses the `cleanWord` function, and stores them in a vector via `push_back()`. My function `sortWords` (`word.begin`, `words.end`) here sorts the words alphabetically. Next, the function `getWordCounts` is something that I used, which is stored as vector pair values. If the words are empty, I just end the function by returning early. I use a if statement nested in a for loop to figure out if the next word in the vector is the same as the previous word, which will increase the count by one. Once there are no more of that word, then I `push_back()` the current word and the count via `make_pair`, and this does not run into issues with there being further words later on in the sequence in the vector because of the alphabetical order. All of these pairs are stored in the vector `counts`. `Int main()` here runs `ifstream` to read the file "holmes.txt", defines the vector `words`, and runs all previous functions to then use a for loop and the `.first` and `.second` functions to print out the first and second items in the pairs (the words and their counts). I then close the file and return 0.